

Chemistry of bleaching powder :-

Bleaching powder, also known as calcium hypochlorite, is a chemical compound with the formula $\text{Ca}(\text{ClO})_2$.

It is a strong disinfectant and bleaching agent.

Composition :-

Bleaching powder is composed of :

- 1- calcium hypochlorite ($\text{Ca}(\text{ClO})_2$) : 30-35%.
- 2- calcium hydroxide ($\text{Ca}(\text{OH})_2$) : 30-40%.
- 3- calcium chloride (CaCl_2) : 10-20%.
- 4- water (H_2O) : 10-20%.

Preparation :-

Bleaching powder is prepared by reacting chlorine gas with slaked lime (calcium hydroxide).



Properties :-

- 1- Bleaching agent :- calcium hypochlorite is a strong oxidizing agent that breaks down chromophores, resulting in the removal

of color.

2- Disinfectant: Bleaching powder is effective against a wide range of microorganisms, including bacteria, viruses and fungi.

3- pH: Bleaching has a high pH (12-13) making it a strong alkaline substance.

Uses:-

1- Laundry: Bleaching clothes and removing stains.

2- water treatment: Disinfecting and purifying water.

3- Swimming pools: Maintaining proper pH and disinfecting pool water.

4- Food industry: Sanitizing equipment and surfaces.

5- Medical: Disinfecting medical instruments and surfaces.

Shampoo :-

Shampoo is a hair product specifically formulated to cleanse the hair and scalp.

Ingredients :-

Surfactants: Cleansing agent that lower the surface tension of water allowing it to spread and penetrate more easily.

e.g. Sodium lauryl sulfate (SLS) helps remove dirt and oil from hair.

Thickeners: Ingredients that increase the viscosity of a formulation. e.g.

Xanthan gum is often used in shampoos to create a desirable texture.

Preservatives: Chemical used to prevent microbial growth. e.g.

Parabens inhibit the growth of bacteria and mold.

Fragrances: Substance added to give a pleasant scent. e.g.

Essential oils like lavender or citrus provide fragrance.

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Colourants: Dyes or pigments added.

e.g. Dand c yellow No. 5 is used to give Shampoo a yellow color-hue.

Function of Shampoo :-

- wash hair effectively.
- Remove dirt completely.
- Not have any side effects or irritation.
- To Prevent dandruff.

pH:-

- pH is a scale measurement of the acidity or basicity.
- Human hair has pH balance between 4.5 and 5.5.
- The pH of Shampoo usually ranges from 5 to 11.
- If the pH is too acidic, it will cause the cuticle damage and pores can may open.

Chemistry of Ceramics :-

A ceramic is an inorganic non-metallic solid, generally based on an oxide, nitride, boride or carbide that is fired at high temperature.

Ceramics may be glazed prior to firing to produce a coating that reduces porosity and has smooth, often colored surface.

Raw materials :-

Raw materials for ceramics include clay, kaolinite, aluminium oxide, silicon carbide, tungsten carbide and certain pure elements.

Properties of Ceramics :-

- i- high hardness
- ii- High melting points
- iii- poor electrical and thermal conductivity
- iv- low ductility
- v- chemical resistance.

Types :-

1- Earthenware :- Made from clay and other natural materials, often used for pottery and tableware.

Stoneware :- Made from combination of

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clay and other minerals, often used for cookware and decorative items.

Porcelain: Made from a combination of clay, feldspar and quartz, often used for fine china and decorative items.

Advanced ceramics: Made from a variety of materials including alumina, zirconia and silicon carbide, often used for industrial and chemical applications.

Applications:

Tableware and cookware: Ceramics are used to make plates, bowls, cups and cooking vessels.

Sanitary ware: Ceramics are used in toilets, sinks and bathroom fixtures.

Bricks & Tiles: Ceramics are used in construction of building walls, floors, roofs etc.

Medical: Ceramics are used in implants. Such as hip and knee replacements, dental implants and surgical instruments.

Nuclear: Ceramics are used in nuclear

reactors and fuels due to high ^{the} radiation resistance.

Biotechnology: ceramics are used in bio-reactors and tissue engineering scaffolds due to their high biocompatibility.

Textiles: Ceramics are used in textile manufacturing, such as in ceramic-coated fabrics and fibers.

Chemical Reactions:

During the ceramic manufacturing process various chemical reactions occur including:

1- Decomposition: Raw materials decompose into simpler compounds, releasing gases like water vapours and CO_2 .

2- Solid-state: Particles react with each other to form new compounds such as silicates and aluminates.

3- Vitrification: The formation of glassy phase, which can occur through the reaction of silica and alumina.

4- Crystallization: The formation of crystalline phases, such as quartz and feldspar.